

Pooja Yogesh Athawale

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SUMMARY

Artificial Intelligence & Data Science Professional (M.S. Data Science) with Production-Grade experience building AI-driven solutions across Research and Enterprise Environments. Skilled in Python, PyTorch, RAG architectures, Vector Databases, LangGraph, Gemini API, and Vertex AI on GCP and AWS. Delivered 40% Reporting Time Reduction and 25% Cloud Cost savings at Enterprise Scale. Targeting forward-Deployed AI Engineering roles bridging frontier AI and enterprise production systems.

TECHNICAL SKILLS

Artificial Intelligence, Machine Learning & Deep Learning: PyTorch, Scikit-learn, VAE, Generative AI, LLM Fundamentals, Prompt Engineering, Hugging Face, NLP, Computer Vision, Feature Engineering, Predictive Modeling

Agentic & GenAI: RAG, Vector Databases, LangGraph, Multi-Agent Systems, Gemini API, Vertex AI, Google AI Studio, AI-Driven Solution Development

Cloud & Infra: GCP, AWS (S3, RDS, DynamoDB, Lambda), Databricks, PySpark, ETL Pipelines, APIs & System Integration

Programming: Python (Pandas, NumPy), SQL, R, PostgreSQL, SQLite, OOP

Analytics & Viz: Tableau, Power BI, EDA, KPI Dashboards, Matplotlib, Seaborn, Business Intelligence

Tools: GitHub Copilot, Jupyter, Jira, Confluence, Agile/Scrum, Cursor, Claude

HONORS, AWARDS & PRESENTATIONS

3rd Place – Arkansas Academy of Science (AAS) Conference 2026, Southern Arkansas University: Variational Autoencoder-based Breast Tumor Analysis Research.

Research Poster – Developing and Advancing Research Together (DART) Poster Competition, Little Rock, AR, Tumor Representation and Reconstruction Results.

Research Paper Accepted – Intelligent Systems Conference (IntelliSys) 2026, Amsterdam, The Netherlands, **Deep Generative Modeling** applied to Medical Image Analysis.

EXPERIENCE

Research Assistant, Southern Arkansas University, Magnolia, AR

May 2025 – Jul 2026

Designed and trained a production-grade VAE in PyTorch on 780 breast ultrasound images, achieving 15–20% reconstruction loss reduction; built end-to-end evaluation pipelines that cut model iteration cycles by 35% through structured experiment tracking and observability frameworks. Optimized latent space via 8+ dimension benchmarks, settling on 32-D encoding with 22% fidelity gain and 4× compression; applied Generative AI and LLM fundamentals to contextualize VAE research within LLM-era representation learning resulting in an accepted paper at IntelliSys 2026. Mentored cross-functional research peers across 3 active projects, influencing experimental design and improving team documentation quality by 40% as part of a Master's-level AI research initiative.

Data Analyst & Cloud Specialist, Hiten Technologies, Pune, India

Jan 2018 – Nov 2020

Engineered production-grade ETL pipelines in Python & SQL ingesting structured and unstructured data, reducing manual reporting time by 40% and improving enterprise data accuracy by 30%; architected AWS workflows at 99.9% uptime with 25% infrastructure cost reduction. Developed 6 executive KPI dashboards in Tableau & Power BI with real-time REST API integration, enabling stakeholder teams to make data-driven decisions 3× faster; led technical discovery sessions translating needs into 12 automated pipeline specs, cutting clarification cycles by 50%. Applied predictive modeling across 4 forecasting initiatives, improving accuracy by 18% and identifying cost-saving trends that influenced \$200K+ in annual operational budget decisions.

PROJECTS

BankAI Customer Decision Prediction – Evaluated 6 supervised learning algorithms (Logistic Regression, SVM, Decision Trees, Random Forest, ANN, CNN) with full evaluation framework (Accuracy, Precision, Recall, F1, ROC-AUC); engineered complete modeling pipeline with preprocessing, feature engineering, and class-balance handling.

AuroraPredict AI Data Mining & Predictive Modeling – Built data-mining and predictive modeling pipeline on 10+ years of real-world solar/light-pollution data with end-to-end feature engineering, imputation, and long-term trend analysis. **End-to-**

End Data Analytics Pipeline in Databricks – Designed reusable PySpark ETL pipelines covering ingestion, transformation, validation, and notebook-based visualizations for reproducible analytics workflows.

EDUCATION

Master's in Data Science, Southern Arkansas University, Magnolia, AR

Jan 2025 – Jul 2026

Post Graduate Program in Cloud Computing, University of Texas at Austin, Austin, TX

Bachelor's in Information Technology, Dr. Babasaheb Ambedkar Marathwada University, India

CERTIFICATIONS

Google AI Professional Certificate | AWS Machine Learning Foundation | AWS Cloud Practitioner | Salesforce AI Associate

Tableau Desktop Specialist | Scrum Fundamentals | Six Sigma Yellow Belt | Jira & Confluence

In Progress – Google Advanced Data Analytics Professional Certificate | Google Business Intelligence Professional Certificate